

Trading Ahead of Treasury Auctions

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ABSTRACT

I develop a model explaining the gradual price decrease observed ahead of anticipated asset sales, such as Treasury auctions. In the model, risk-averse investors expect an increase in the net supply of a risky asset, about which they have a noisy signal. They face a trade-off between hedging the noise with long positions, and speculating with short positions. As a result of hedging, the equilibrium price is above the expected price. As the sale approaches, the noise decreases due to the arrival of information, investors hedge less, and the price decreases. I illustrate the relevance of the theory in the days preceding Italian Treasury issuances. I find that meetings between the Treasury and primary dealers explain a 2.4 bps yield increase.

JEL classification: G11, G12, E43.

Keywords: anticipated supply shocks; supply risk; Treasury auctions; market making

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I. Introduction

While the asset pricing literature has studied prices following large asset sales such as Treasury auctions, little is known about prices preceding a sale. In this paper, I highlight a new channel through which anticipated asset sales affect prices in a predictable fashion.

More precisely, I develop a theory explaining why bond prices decrease gradually in a predictable fashion ahead of Treasury auctions. I show that these price movements arise when investors *imperfectly* anticipate the extent to which the presence of natural buyers will push the price up on the day of the sale. As a result, the signal about the *net* supply shock is noisy, and this situation triggers a *positive* hedging demand from those who expect to profit from a low sale price. The hedging demand gradually declines when the noise of the signal gradually decreases. Hence, the price decreases gradually. I illustrate the empirical relevance of the theory in the days leading up to Italian government bond issuances. I use meetings between the Treasury and primary dealers as events that reduce the noise of the signal about the net supply shock. I find that the meetings, together with other relevant events, explain a circa 15 bps drop in Treasury prices. The findings cannot be explained by the on/off-the-run effect since that this paper's empirical setting is free from this effect. I also discuss applications to other instances of anticipated asset sales, such as Seasoned Equity Offerings (SEOs) and the rebalancing of future contracts.

The empirical literature has found that, in the few days leading up to US Treasury auctions, the price of current issues decreases gradually, reaches a minimum on auction day, and then increases gradually. The pre-auction price decline has traditionally not been taken into account when computing the issuance cost born by the Treasury. Given the pre-auction decline, this cost could be up to ten times larger than previously thought (Lou, Yan and Zhang (2013)).

The decreasing price pattern prior to an auction is a puzzle in that it implies that some investors are willing to buy bonds ahead of the auction at a price which, on average, exceeds the auction price. Admittedly, impatient buyers would be ready to buy at a premium before the auction, rather than to wait and buy on the auction day at a noisy price. But conversely, impatient sellers would be ready to sell at a discount. As a result, one would have expected the pre-auction price to be equal to the expected price.

Existing explanations of the price pattern are not entirely satisfying, for two reasons. First,

some theories do not address the question of why investors choose to buy ahead of the auction at the high price. Indeed, some papers take this behavior as exogenous (as in the limited risk-bearing channel mentioned in Lou, Yan and Zhang (2013), and Fleming and Liu (2017)), while others use inattention in order to generate buying ahead of the auction (Duffie (2010)). Second, theories based on the on/off-the-run effect cannot explain why the price pattern also exists when there is no on/off-the-run effect. Therefore, there is a need for a theory which rationalizes buying ahead of large anticipated sales, when the buyers are fully attentive, and when the on/off-the-run effect does not play a role.

The model's key insight is that investors have a *positive* hedging demand ahead of the sale. That is, they buy (or do not sell as much as otherwise) securities which are positively correlated with the to-be-sold security. The sign of the hedging demand is crucial for this paper's mechanism because the hedging demand keeps pre-shock prices *above* the price expected on the shock date. In the context of Treasury auctions, the reasons of this positive hedging demand are as follows. In Treasury auctions, the date and the size of the auction may be perfectly anticipated, but not necessarily the presence of natural buyers on the auction day (e.g. insurance companies or mutual funds). In case of a higher-than-expected presence of natural buyers, the Treasury can sell the bonds without relying much on liquidity providers (e.g. primary dealers). This represents an income loss for the liquidity providers, in the form of a lower price discount conceded by the Treasury. If this risk materializes on the auction day, the liquidity providers can recoup their income losses by selling to natural buyers assets that are in high demand, i.e. assets positively correlated with the auctioned bond. Hence, there is a positive hedging demand for these assets prior to the auction. Interestingly, this line of reasoning can be applied to non-Treasury contexts as it does not crucially depend on the institutional details of Treasury auctions. In particular, in virtually any market, a large seller is obliged to propose a discount, so that other investors agree to absorb the supply. And the size of this discount is not necessarily known in advance.

It is important to study the gradual price decrease ahead of auctions for three reasons. First, sovereign bonds are core securities: they are a funding source for Treasury departments, are used as collateral in the Repo market, and have large derivative markets. Therefore, understanding their prices is important, especially around frequent events such as Treasury auctions. Second, the empirical literature has documented, in non-Treasury contexts, a pattern which is similar to

that occurring around Treasury auctions; e.g. around SEOs (Corwin (2003); Meidan (2005)), and around anticipated rebalancing of future contracts (Bessembinder, Carrion, Tuttle and Venkataraman (2016)). Hence, what happens around Treasury auctions can shed some light on what happens in these markets as well. In that respect, this paper's model is general, and can be adapted to fit the institutional details of these contexts. Third, there is a theoretical interest to the gradual price decrease ahead of auctions. Indeed, while the theoretical literature has studied prices following asset trades, little is known about prices before a trade. Moreover, these theories of *post-trade* prices do not readily apply to *pre-trade* prices.¹

In the baseline model, there are three dates, risk-averse investors (e.g. liquidity providers such as primary dealers), a risky asset (e.g. a Treasury bond), and a riskless asset used as numeraire. In the third date, the assets pay off. In the second date, there is an auction modeled as a change in the net supply of the risky asset. In the first date, the investors trade while having a noisy common signal about the change in net supply (*net supply noise*). The net supply is simply defined as the supply minus the quantity of risky asset bought by unmodeled natural buyers.

First and foremost, I find that investors seek to hedge the net supply noise with long positions in the first date. Indeed, a long position between the first and second dates will have a high return precisely when the return between the second and third dates is low, i.e. in the scenario where the net supply is small.

Second, I find that investors also take short positions to speculate on the price difference between the first and the second dates: they seek to sell at a high price in the first date, and to close their positions at a low price in the second date. Therefore, there is a trade-off between short speculative, and long hedging positions. Third, I find that the equilibrium price is above the expected sale price as a result of the hedging demand. Fourth, as the net supply noise decreases, the hedging positions decrease relatively to the speculative positions, and thus the price decreases. This last result is generated through comparative statics, but is robust to using a fully dynamic model featuring an intermediate date ("Date 1.5") where the noise of the shock is scheduled to decrease (cf. Appendix).

Then, I develop two extensions of the model in order to match additional empirical observations. The first extension generates a post-sale price increase, while the second extension generates short-

¹For instance, Grossman and Miller (1988) assume that security dealers are as likely to buy as they are to sell, which does not apply to issuances.

selling in equilibrium. In the first extension, I introduce an additional date (“Date 2.5”) where the size of the net supply is scheduled to decrease, but the size of the decrease is not perfectly known in advance. I find that, on average, the price decreases from Date 1 to Date 2, and increases from Date 2 to Date 2.5. In the second extension, some investors (“short-term”) exit the market in Date 2, and are therefore not exposed to net supply noise. I find that the short-term investors are short-selling the risky asset in Date 1.

Overall, the price dynamics predicted by the model are consistent with the v-shaped price patterns documented around Treasury auctions (Lou, Yan and Zhang (2013); Beetsma, Giuliodori, de Jong and Widiyanto (2016); Keloharju, Malkamaki, Nyborg and Rydqvist (2002)), as well as around other anticipated sales, such as seasoned equity offerings (SEOs) (Corwin (2003); Meidan (2005)), or the rebalancing of future contracts (Bessembinder, Carrion, Tuttle and Venkataraman (2016)). It is also consistent with the documented short-selling activity around auctions (see e.g. Keane (1996)).

Next, I derive an empirical implication: upon the *scheduled* release of information about the net supply, the price moves more if the information is negative (i.e. if it indicates a larger-than-expected net supply) than if positive (i.e. if it indicates a smaller-than-expected net supply). An example of such information is the scheduled release of an expert report about what will be the demand for the asset on the auction day. The asymmetry between positive and negative information is due to the fact that the release creates a reduction in noise, which in turns exerts a downward price pressure. The downward price pressure dampens the price increase which trivially results from positive information, and it amplifies the price decrease which results from negative information. Hence, the price moves more if the information is negative. In particular, in a sample that is not systematically biased towards positive nor negative information, the model implies an average price decrease upon a release of information.

Finally, I illustrate the empirical relevance of the model’s implication in the days preceding Italian government bond issuances. The sample consists of 849 auctions of Italian Treasuries from 2000 to 2015. I use the fact that Italian sovereign bonds are typically issued over several consecutive months in a staggered fashion, thus making available to the econometrician the price of a given bond prior to the auction of this very same bond. In particular, the on-the-run premium (Krishnamurthy (2002)) is negligible compared to a setting that would use the pre-auction price of the old bond

in combination with the post-auction price of the new bond. I exploit two scheduled pre-auction events: first, private meetings between the Italian Treasury and primary dealers where both parties discuss the upcoming auctions; second, announcements of the issuance size by the Treasury. Both, the dealers' meeting and the size announcement are shown to be informative, therefore reducing the noise of the signal about the net supply. Finally, I use exogenous variations in the timing of these meetings in order to control for competing mechanisms. As predicted, I find that the yield of the reopened bond increases by 2.4 bps more on these two occasions than on non-information days. The effect is equivalent to a circa 15 bps drop in prices, and is around three times larger than the half-spread charged by dealers on the secondary market. Furthermore, in a back-of-the-envelope calculation, I find that this paper's mechanism explains 77% of the pre-auction pattern.

This paper contributes to the theoretical literature on anticipated supply shocks by highlighting a new link between these shocks and price predictability. It is the first paper in predicting that a gradual decrease in the noise about a supply shock creates a gradual and predictable price decrease. The literature has identified four theoretical channels through which the price ahead of a supply shock is above the sale price. The first channel is highlighted in Bessembinder et. al. (2016), where traders sell ahead of the sale, thus depressing the price. In the same channel, Fardeau (2016) shows that arbitrageurs buy ahead of the sale so as to reduce the price impact incurred on the date of the sale. In a second channel, Beetsma, Giuliodori, de Jong and Widiyanto (2016) use a model à la Ho and Stoll (1983), and show that the market maker sets the price higher before the auction because her risk exposure is lower than right after the auction. My mechanism is different from these two channels as I rely on the noise of the supply shock, whereas the supply shock is perfectly anticipated in the first channel, and the second model is static. In addition, contrary to Bessembinder et. al. (2016) and Beetsma et. al. (2016), I rationalize pre-auction buying, and I have predictions regarding the gradual price decrease. And, contrary to Fardeau (2016), I illustrate the empirical relevance of the model.²

A third channel is highlighted in Duffie (2010), where the price ahead of an anticipated sale is above the expected price, and decreases gradually as inattentive investors arrive gradually in the market (see also Bogousslavsky (2016), among others). Both, my paper and Duffie (2010), generate

²Fardeau (2016) has a *risky arbitrage extension* where the supply shock is imperfectly anticipated, like in my model. He predicts that a decrease in the noise of the supply shock would create a price increase, whereas I predict the opposite.

the pre-auction price decrease, and the post-auction price increase. They both assume that there is not enough capital to immediately absorb the supply shock. However, I relax Duffie (2010)'s assumption that some investors who trade *before* the auction cannot trade *on* the auction date. Hence, in my model, investors choose to buy the bond ahead of the auction at a high price, even though they would be able to buy it at a lower average price on the auction date. In exchange for relaxing this friction, my model assumes that pre-auction prices are not perfectly informative about the demand of the natural buyers. Therefore, the two papers have different mechanisms, and the setting used in this paper's empirical section allows to control for Duffie (2010)'s mechanism.

This paper belongs to a fourth channel. By focusing on imperfectly anticipated supply shocks, my model is close to Vayanos and Wang (2012), Admanti and Pfleiderer (1991) and Vayanos and Woolley (2013). In Vayanos and Wang (2012), the shock is imperfectly anticipated, like in my model. However, in their paper, the shock has a zero mean. As a result, Vayanos and Wang (2012) do not predict that the price ahead of the shock is above the expected price (see also Lo, Mamaysky, and Wang (2004), and Huang and Wang (2009, 2010)). This paper is also close to the sunshine trading literature and, specifically, the model in Admanti and Pfleiderer (1991). In both models, there are some liquidity providers who are uncertain about the demand of liquidity traders. In both models, this uncertainty is elicited through an announcement. However, the demand of liquidity traders in Admanti and Pfleiderer (1991) has a zero mean. So, unlike my model, they do not predict a price decrease upon the announcement. My paper is closer to Vayanos and Woolley (2013) since that both papers feature an imperfectly anticipated shock with a non-zero mean. Hence, they both predict that the price ahead of the supply shock is above the expected price due to an endogenous hedging demand. Importantly though, Vayanos and Woolley (2013) do not formulate a prediction linking the price level to the noise level of the signal. This prediction is key to generate the gradual price decrease observed in the empirical literature. Therefore, my largest contribution with respect to their paper is to make that prediction. Another important contribution with respect to their paper is that I illustrate the empirical relevance of the channel.

Finally, the paper contributes to the empirical literature on anticipated supply and demand shocks by investigating the drivers of the price dynamics ahead of these shocks. Lou, Yan and Zhang (2013) find that the pre-auction price decrease is stronger in times when Treasury interest rates are more volatile, when it is more costly for dealers to obtain leverage, and when the auction

size is larger. They argue that these findings are consistent with their hypothesis that dealers sell in advance their auction participation due to limited risk capacity, thus depressing prices ahead of the auction. (See also Fleming and Liu (2017) who find that the magnitude of the price decrease is positively correlated with bid-ask spreads, price impact, and volatility). Importantly, these papers' findings do not speak against the net supply noise story developed in my paper. In particular, the volatility-based proxies, used in those papers to capture dealers' constraints, are likely to capture net supply noise as well. The biggest empirical difference between my paper and these papers is that I focus on the drivers of the time series within a given auction, not on the drivers of the cross-section. And my findings are not predicted by the dealers' constraints narrative. Another difference is that I exploit staggered auctions, and remove from my sample all bonds that are about to go off-the-run; thus allowing to more clearly reject the possibility that the pattern is driven by the on/off-the-run phenomenon (Krishnamurthy (2002)).

This paper is also related to Fleming and Rosenberg (2008). They study how primary dealers manage their inventories, in particular around auctions. They find that primary dealers are compensated for holding inventory, as the price at which they buy is lower than on adjacent weeks. They also find that primary dealers' inventory in coupon-bonds is on average negative and do not find any inventory adjustment on the weeks prior or after an auction. My paper is compatible with their findings. Like them, I find that the price is lower on auction day than on adjacent weeks. In addition, this paper's model allows for negative inventories and does not necessarily predict inventory –but price– adjustments. Nonetheless, the two papers differ in their research questions. In particular, unlike them, I study the mechanism that leads the pre-auction price to stay above the auction price.³

The rest of this paper is organized as follows. Section 2 develops the model. Section 3 illustrates the empirical relevance of the model. Section 4 discusses the results. Section 5 concludes.

³This paper also relates to the Treasury auction literature. While this paper mainly features net supply noise and the existence of a pre-auction market, some papers focus on other features, such as strategic bidding behaviors (e.g. Cammack (1991); Nyborg, Rydqvist, and Sundaresan (2002)), information in the when-issued market (e.g. Nyborg, and Sundaresan (1996)), the obligation of primary dealers to participate (e.g. Fleming, Rosenberg and Joshua (2007)), primary dealers' private information (Boyarchenko, Lucca and Veldkamp (2016)), or auction design (e.g. Back and Zender (1993)).

II. Model

A. Objective of the model

I build a model that aims to rationalize why Treasury bond prices have been documented to decrease gradually ahead of auctions. Figure 1 offers a graphical representation of the pattern around auctions. Figure 1 uses an empirical setting which is presented in Section III. What Figure 1 shows is a *slow* price decrease *before the sale*, followed by a slow increase. Instead, what is well accepted in the literature is that a large sale entails an *immediate* price decrease *at the time of the sale*, followed by a slow increase. The surprising part is therefore the pre-auction price dynamics, which is the focus of this paper.

The model is a three-date portfolio management model, with a change in the net supply of the risky asset in the intermediate date. There are two sets of crucial assumptions in the model. First, the change in net supply in the intermediate date is imperfectly known in advance by the investors, and has a positive mean. Second, investors are risk-averse and have a long-term horizon.

B. Set-up

There are three dates ($t = 1, 2, 3$), a riskless asset and a risky asset. The riskless asset is the numeraire.

At $t=1$, the net supply of the risky assets is $\bar{\theta}$. **At $t=2$** , the net supply of the risky asset is increased by an amount Z . Z is known at $t=2$, while only its distribution is known at $t=1$, with $Z \sim N(\bar{Z}; \sigma_Z^2)$, $\bar{Z} > 0$.⁴ Equivalently, Z can be decomposed into a common signal $\bar{Z}$ and a common error term with distribution $N(0; \sigma_Z^2)$. As such, σ_Z^2 is a measure of the noise of the common signal about Z . **At $t=3$** , the risky asset pays off D units. D is known at $t=3$, while its distribution is known at $t=1$ and $t=2$, with $D \sim N(\bar{D}; \sigma^2)$. D and Z are assumed to be independent.

There is a measure one of investors who trade the assets **at $t=1$ and $t=2$** . Their utility function is:

⁴Note that when $\bar{\theta}$ is negative, the assumption $\bar{Z} > 0$ is not enough. The assumption $\bar{Z} > -\bar{\theta}$ is needed

$$U(W_3) = -\exp\{-\alpha W_3\} \quad (1)$$

where W_3 is the individual's wealth in $t=3$, and $\alpha > 0$ is the coefficient of absolute risk aversion. Investors have an endowment C_0 in the riskless asset, and θ_0 in the risky asset at the start of $t=1$.

Figure 2 illustrates the timing of the model.

C. Discussion of the assumptions

Table I illustrates the interpretation of the model in the context of Treasury auctions. Investors are primary dealers with an initial inventory of $\bar{\theta}$. To accommodate for the empirical evidence in Fleming and Rosenberg (2008), $\bar{\theta}$ can be negative.⁵ The primary dealers absorb a quantity Z of assets at $t=2$, the auction day. Z is the share of the new issue minus what was not sold to “natural buyers” (e.g. foreign funds, investment funds, individuals).

Why is it appropriate to assume that the change in inventory Z is noisy at $t=1$? According to Fleming (2007), 40% of long-term US bond volume is bought by non-primary dealers, including by foreign investors, and investment funds. Crucially, the share of non-primary participants has varied from 0% to 67% in the United States (Fleming (2007)). Given this large variation in the demand of non-primary dealers, it may be challenging to perfectly predict this demand. Indeed, some of these non-primary bidders bid directly at the auction. Hence, their demand is only known after the auction. In addition, the demand of those who bid through a primary dealer remains noisy until the primary dealer has effectively collected all her clients' orders. Even then, a given primary dealer only receives an imperfect signal of the overall demand, as each primary dealer collects a fraction of the total orders (Boyarchenko, Lucca and Veldkamp (2016)).

Finally, I justify why Z is assumed to have a strictly positive average. Alternatively, one could have assumed Z to have a zero mean. Whether Z is positive or null depends on whether we assume the issuance to have a price impact, or not. If the issuance was unpredictable, there

⁵In that case, $\bar{\theta}$ can no longer be interpreted as the asset's initial net supply in the context of Treasuries, but can rather be interpreted as the holdings of the investors present in the market

would necessarily be an impact (Grossman and Miller (1988)). However, issuances are predictable events. As studied in the sunshine trading literature, this predictability is a mechanism for the issuer to signal its lack of superior information, and gives the issuer time to attract natural buyers on the auction day (Admanti and Pfleiderer (1991)). By assuming Z to have a positive mean, the model assumes that the natural demand on auction day is insufficient on average to absorb the entire issue size. This insufficient natural demand is supported by empirical evidence which shows that the inventory of Treasury dealers increases on average after auctions (Fleming and Rosenberg (2008)).

D. Solution

I start by deriving the equilibrium at $t=2$. The results at $t=2$ are standard. The price of the risky asset at time t is denoted P_t .

Investors maximize

$$\mathbb{E}_D \left[- \exp \left\{ - \alpha (\theta_2 D + C_0 - (\theta_1 - \theta_0) P_1 - (\theta_2 - \theta_1) P_2) \right\} \right] \quad (2)$$

i.e. the expectation over the risky pay-off D of the wealth at $t=3$ i.e. the value $\theta_2 D$ of the total risky portfolio at $t=3$, plus the endowment in cash C_0 , minus the cost $(\theta_1 - \theta_0) P_1$ of the additional risky position taken at $t=1$, minus the cost $(\theta_2 - \theta_1) P_2$ of the additional risky position taken at $t=2$.

The optimal demand function for the risky asset at $t=2$ of investors is

$$\theta_2^*(P_2) = \frac{\bar{D} - P_2}{\alpha \sigma^2} \quad (3)$$

where θ_2 is the investors' total holdings at $t=2$.

The market clearing condition is

$$\bar{\theta} + Z = \theta_2^*(P_2^*) \quad (4)$$

Using (3) and (4), I show that the equilibrium price for the asset at $t=2$ is

$$P_2^* = \bar{D} - \alpha\sigma^2(\bar{\theta} + Z) \quad (5)$$

The equilibrium holdings at $t=2$ is

$$\theta_{2,i}^* = \bar{\theta} + Z \quad (6)$$

Finally, using (2), (5) and (6), I obtain the value function at $t=2$, and Lemma 1. Lemma 1 is useful to get an early intuition about the model's mechanism, later in this section.

Lemma 1: *Investors' value function at $t=2$ is a function of Z , decreasing over $(-\infty; \theta_1 - \bar{\theta}]$, increasing over $[\theta_1 - \bar{\theta}; +\infty)$, and concave over $[\theta_1 - \bar{\theta} + \frac{1}{\alpha\sigma}; +\infty)$:*

$$V_2(Z, W_2) = -\exp - \left\{ \alpha \left(W_2(Z) + \frac{1}{2} \alpha \sigma^2 (\bar{\theta} + Z)^2 \right) \right\} \quad (7)$$

where $W_2(Z) = C_0 - (\theta_1 - \theta_0)P_1 + \theta_1 P_2(Z)$

The interpretation of Lemma 1 is the following. The decreasing and increasing monotonicity indicates that the investors' expected utilities are increasing in the absolute size of the net supply. In addition, the concavity of the value function –in the region where the function is increasing in Z – suggests that the investors will try to avoid situations where the net supply turns out to be *smaller* than expected. In exchange, they are willing to forego the extra expected utility derived from a situation where the net supply turns out to be *larger* than expected. Overall, Lemma 1 suggests that investors will try to hedge at $t=1$ the possibility that Z turns out to be smaller than $\bar{Z}$.

I now derive the demand function and the equilibrium price at $t=1$. This derivation leads to Proposition 1, the model's most important result.

The investors' problem consists in maximizing the expectation over Z of the value function given

in Equation (7). More precisely, they choose θ_1 such as to maximize the following:

$$\mathbb{E}_Z \left[-exp - \left\{ \alpha \left(W_1 + \theta_1 \left(\bar{D} - \alpha \sigma^2 (\bar{\theta} + Z) - P_1 \right) + \frac{1}{2} \alpha \sigma^2 (\bar{\theta} + Z)^2 \right) \right\} \right] \quad (8)$$

where $W_1 = C_0 + \theta_0 P_1$

Proposition 1: *Investors' demand function for the risky asset at $t=1$ is composed of a speculative demand, and a positive hedging demand:*

$$\theta_1^*(P_1) = -\frac{P_1 - \mathbb{E}_Z(P_2)}{\alpha Var_Z(P_2)/(1 + \alpha^2 \sigma^2 \sigma_Z^2)} + \alpha \sigma^2 (\bar{\theta} + \bar{Z}) \frac{Cov(P_2, -Z)}{Var_Z(P_2)} \quad (9)$$

where $\alpha \sigma^2 \frac{Cov(P_2, -Z)}{Var_Z(P_2)} = Corr(P_2, -Z) = 1$

Equation 9 offers an intuitive decomposition of the demand function. The first term, including the negative sign in front of it, corresponds to a speculative demand: given that investors expect the price at $t=2$ to be low due to the supply effect, they seek to sell at $t=1$ and buy back cheap at $t=2$. This trading strategy can also be found in Bessembinder et. al. (2016). The second term is a hedging demand because it depends on the covariance of P_2 with Z . The hedging demand translates into a positive demand for the risky asset.

The economic interpretation of the hedging demand is the following. The sale constitutes an investment opportunity for investors because they can buy cheap at $t=2$ and sell back at $t=3$. Net supply Z is a state variable which determines how profitable this opportunity is. Hence, risk-averse investors seek to hedge the changes in the profitability of this investment opportunity. They do so with an investment which negatively correlates to the state variable Z . In that regard, a long position in the risky asset is valuable because the return of that investment is high when Z is low. Overall, although investors have an incentive to sell the asset at $t=1$ and buy back cheap at $t=2$, they do less of it because they have a motive to invest in the asset. There is therefore a trade-off between short-term profits, and hedging the uncertainty of future profits.

In order to have a better understanding of the demand function, I study some comparative statics. First, the absolute value of the speculative demand decreases in σ_Z^2 . Indeed, the noise

about net supply Z represents a cost for risk-averse investors: the higher σ_Z^2 , the less willing they are to speculate. Second, the higher $\frac{Cov(P_2, -Z)}{Var_Z(P_2)}$, the higher the hedging demand. This is because $\frac{Cov(P_2, -Z)}{Var_Z(P_2)}$ is the “beta” of P_2 with respect to $-Z$: the higher the beta, the better the investors are insured against the risk of Z . Third, the larger $\bar{Z}$, the larger the hedging demand. Indeed, after simplification, the hedging demand is equal to $\bar{Z}$. This means that investors seek to buy in advance at $t=1$ what they otherwise expect to buy at $t=2$. Finally, the level of hedging demand (relatively to speculative demand) increases in σ_Z^2 .

Note that one gets a useful alternative expression for the investors’ demand by replacing in Equation 9 the expression of $\mathbb{E}_Z(P_2)$, $Var_Z(P_2)$ and $Cov(P_2, Z)$:

$$\theta_1^*(P_1) = \frac{1 + \alpha^2 \sigma^2 \sigma_Z^2}{\alpha^3 \sigma^4 \sigma_Z^2} (\mathbb{E}_Z(P_2) - P_1) + \mathbb{E}_Z(\theta_2^*) \quad (10)$$

Using (10) and noting that $\theta_1^* = \bar{\theta}$ is a market clearing condition, I obtain the equilibrium price at $t=1$ and Proposition 2. Proposition 2 is useful for the predictions developed in section 2.F.

Proposition 2: *Buying the risky asset at price P_1^* , and selling it at price P_2^* , gives a negative average return. This return is more negative when the noise σ_Z^2 of the signal about the net supply Z is larger:*

$$\mathbb{E}_Z(P_2^*) - P_1^* = -\frac{\alpha^3 \sigma^4 \sigma_Z^2 \bar{Z}}{1 + \alpha^2 \sigma^2 \sigma_Z^2} < 0 \quad (11)$$

and

$$\frac{\partial(\mathbb{E}_Z(P_2) - P_1^*)}{\partial \sigma_Z^2} < 0 \quad (12)$$

Proposition 2 tells us that the average price difference between $t=1$ and $t=2$ increases as the noise σ_Z^2 increases. This is because, as σ_Z^2 increases, speculation is less profitable. Hence, investors increase their hedging positions relatively to their speculative position.

Finally, I propose two useful benchmarks in Lemma 2, in order to further study the equilibrium at $t=1$:

Lemma 2: $\mathbb{E}_Z(P_2) - \alpha \text{Var}_Z(P_2)\bar{\theta} < P_1^* < \bar{D} - \alpha\sigma^2\bar{\theta}$.

The left-hand side expression is the price at $t=1$ in an economy where investors care only about one-period returns (hence, they do not hedge). The right-hand side expression is the price at $t=1$ in an economy where the market does not expect any sale. Therefore, Lemma 2 tells us that the price at $t=1$ would be lower if there were no hedging motive; and the price would be higher if no sale was anticipated.

E. Extensions – Rationalizing the post-auction increase and short-selling

The baseline model generates a pre-auction price above the expected auction price (Proposition 2). However, there are two empirical facts that the baseline model does not address. First, the fact that the post-auction price is, on average, higher than the auction price (Lou, Yan and Zhang (2013)). Second, the existence of short-selling activity around auctions (see e.g. Keane (1996)). Here, I present two extensions which separately address these points.

Extension 1 (Post-sale price). In this extension, I borrow from the market microstructure literature, and I make the net supply temporary (cf. Grossman and Miller (1988)). The extension features an intermediate Date 2.5 where the net supply is expected to decrease by a noisy amount, back to the initial level $\bar{\theta}$. This noisy supply decrease is the symmetric counterpart of the noisy supply increase occurring in Date 2. More precisely, at $t=2.5$, the net supply of the risky asset is changed by a negative amount $-Z_{2.5}$, with distribution $Z_{2.5} \sim N(\bar{Z}; \sigma_Z^2)$. For simplicity, Z and $Z_{2.5}$ are assumed to be independent. Furthermore, it is assumed that $\alpha^2\sigma^2\kappa^2 < 1$ where $\kappa^2 = \frac{\sigma_Z^2}{1+\alpha^2\sigma^2\sigma_Z^2}$. The rest of the setting is identical to the baseline model.

Proposition 3 (derived from Extension 1): *Buying the risky asset at $t=2$, and selling it at $t=2.5$, gives a positive average return. As in the baseline model, buying the risky asset at $t=1$, and selling it at $t=2$ gives a negative average return:*

$$\mathbb{E}_{Z_{2.5}}(P_{2.5}^*) - P_2^* = \alpha^3 \sigma^4 \kappa^2 \bar{Z} > 0 \quad (13)$$

$$\mathbb{E}_Z(P_2^*) - P_1^* = -\alpha^5 \sigma^6 \kappa^4 \bar{Z} < 0 \quad (14)$$

Extension 2 (Short-selling). In practice, there exists an heterogeneity among investors, such as primary dealers. Indeed, they have different inventories, different mix between own trading and client trading, different weight on their primary dealing activities etc. As a result, they will be differently exposed to the noise of the signal about the net supply. In this extension, I model this heterogeneity in a reduced form by supposing that some investors have a short horizon, and others have a long horizon. Investors with a short horizon are dealers who end up having no hedging concerns. More precisely, I suppose a mass δ of short-term investors ("investors A"), and a mass 1 of long-term investors ("investors B"). Investors A exit the market at $t=2$, while investors B exit the market at $t=3$.

Proposition 4 (derived from Extension 2): *At $t=1$, short-term investors A have a negative holding in the risky asset (i.e. they short-sell). Furthermore, short-selling is inversely related to the noise of the signal about the net supply σ_Z^2 :*

$$\theta_{1,A}^* = \frac{-\bar{Z}}{1 + \delta + \alpha^2 \sigma^2 \sigma_Z^2} < 0 \quad (15)$$

and

$$\frac{\partial(\theta_{1,A}^*)}{\partial\sigma_Z^2} > 0 \quad (16)$$

F. Empirical implications

I now formulate the model's empirical implications. Implications 3 and 4 are new to the literature. The mechanism driving the patterns in Implications 1 and 2 is new as well, although the patterns themselves have been predicted in a few other models. For simplicity, the implications are generated using comparative statics. However, as shown in the Appendix, all the results go through with a fully dynamic model featuring an intermediate date $t=1.5$ where the noise σ_Z^2 is scheduled to decrease. Furthermore, note that I call *to-be-issued asset*, any asset with the same fundamental value as that of an asset scheduled to be issued in the near future.

Implication 1: *Suppose that the noise of the signal about the net supply decreases gradually as the auction date approaches. Then, before the issuance, the price of the to-be-issued asset is above the expected issuance price, and it decreases gradually.*

Implication 1 is based on Proposition 2. It rationalizes the empirical fact that, on average, the price of an on-the-run US Treasury bond is higher before the issuance of a new issue than on the issuance day (Lou, Yan and Zhang (2013)).

Implication 2: *Suppose that the noise of the signal about the net supply decreases gradually as the auction date approaches. Then, before the issuance, trading and short-selling volumes of the to-be-issued asset are higher than usual, and they increase gradually.*

Implication 2 is based on Proposition 4. It justifies why special Repo activity (an indicator of short-selling) increases as the auction date of a new issue approaches (Keane (1996)). Sigaux (2017, chapter 1) finds that the demand for short-selling increases prior to an auction, and he rules out superior information.

Implication 3: *Upon the arrival of scheduled information about the net supply, the price moves*

more if the information is negative (i.e. if it indicates larger-than-expected net supply) than if it is positive (i.e. if it indicates a smaller-than-expected net supply)

Implication 3 is based on Proposition 2 and can be explained as follows. Scheduled information refers to information that is expected to be released in the near future. An example of such information is the scheduled release of an expert report about what will be the demand for the asset on the auction day. The asymmetry in Implication 3 is due to the fact that the information creates a reduction in noise, which in turn exerts a downward price pressure. The downward price pressure dampens the price increase which trivially results from positive information, and it amplifies the price decrease which results from negative information. All in all, the decrease in price is larger than the increase. Hence, Implication 3 predicts a larger price movement if the information is negative, than if it is positive.

The following is a useful alternative formulation of Implication 3. “Take a sample of asset returns corresponding to a strategy of buying *to-be-issued assets* before –and selling them after– a release of information about the assets’ net supplies. Suppose that it is reasonable to assume that the sample is not systematically biased towards positive or negative news. Then, the average sample return is negative.” Indeed, suppose that the price systematically decreases (increases) by 1.25 bps (0.75 bps) after the release of a negative (positive) piece of information. If there is no systematic bias, then on average the release of information entails a decrease of 0.25 bps.

Implication 4: *The difference between the pre-auction price and the expected auction price for the to-be-issued asset is smaller when the auction size varies with the demand of natural buyers.*

Implication 4 is based on Proposition 2. In a primary auction of Treasury assets, the size of the issuance is usually fixed and known in advance but the demand from participants (e.g. mutual funds) might not be. This creates uncertainty regarding the provision of liquidity from primary dealers. On the contrary, if the Treasury decides to let the supply vary with the demand observed on auction day, the noise about net supply σ_Z^2 is lower; this leads to a lower price difference between the first date and the intermediate date.

III. Empirical illustration

In this section, I first check that the patterns in the data are similar to the patterns predicted in Implications 1 and 2. Then, I illustrate the empirical relevance of Implication 3. As is conventional in these markets, I use bond yields instead of prices. Since yields move inversely to prices, the decrease in price predicted by the model translates into an increase in yield.

A. Institutional details

I study *reopenings* of Italian sovereign bonds. In Italy, sovereign bonds are typically reopened several times, i.e. they are issued in a staggered fashion, over several consecutive months. For example, a two-year bond is issued four times on average, over four months. Reopenings are not specific to Italy, but few countries apply a staggered issuance process for such a wide array of maturities, and over so many consecutive months. In particular, US reopenings are used for a limited set of bonds (ten- and 30-year bonds), a limited number times per issue (one to two), and do not always take place. One benefit of this setting is that the yield data for the reopened bonds are available to the econometrician prior to the reopening. The reason for this is that these bonds have been traded in the secondary market for a least one month prior to the reopening. Importantly, by comparing the yield of the same bond before and after an auction, one avoids the caveat of comparing the yield of a soon-to-be off-the-run bond with the yield of a fresh on-the-run bond. Doing so would mean capturing the full on-the-run premium given that the premium is a function of the remaining maturity of the on-the-run bond (Goldreich, Hanke and Nath (2005)). Because this paper's main specification consists in comparing the yield of a single bond from one day to the other, the variation in on-the-run premium captured in the paper's main results is negligible.

Figure 3 presents the Italian issuance timeline. Four points are worth mentioning. The first point of interest is the fact that the reopening date can be precisely predicted in most cases by the market by using historical data (see Table II). The second point is the existence of a meeting where the Treasury and the primary dealers share their views about the reopenings of the next two weeks. The third point is the fact that the Treasury announces the size of the reopening two to four days before the issuance (after market close). The final point of interest is that the number of days separating the reopening date from the dealer meetings –and from the auction size announcement–

is heterogenous as it depends on the bond's maturity and on the period (see Table III).

B. Data

The sample is composed of *reopenings* of fixed-rate two to 30-year Italian sovereign bonds from 2000 to 2015. The largest sample is composed of 849 reopenings. The yield data is from Datastream (RY datatype) and correspond to end-of-day dealer quotes. The trading volume data are also from Datastream (VA datatype) and are based on face values. The yield curve data used as control come from Bloomberg. The Repo data come from MTS (Monte Titoli S.p.A). I keep reopenings for which there exists yield data on Datastream, and I drop eight observations owing to data quality. Table IV and Table V report some summary statistics.

C. Are the patterns in the data similar to the patterns in Implications 1 and 2?

In this part, I study the price and volume patterns present in the data. I check that these patterns are similar to the patterns predicted in Implications 1 and 2. In detail, for each t in $(-5, +5) \setminus \{0\}$, I test the null hypothesis $\alpha_t = 0$ in the following t-test specification:

$$X_{n,d-t} - X_{n,d} = \alpha_t + \epsilon_{n,d-t} \quad (17)$$

where t is the number of days before or after the reopening date of bond n , d denotes the reopening date, and $d - t$ is the observation date. $X_{n,d-t}$ denotes a relevant market variable (secondary yield, log of secondary trading volume, or log of special repo volume) for the reopened bond n on date $d - t$. The clustering is done at the year-month-half level ("bi-weekly"), i.e. for each month of each year, two clusters are created: one for the first half, and one for the second half of the month.⁶

Table VII reports the point estimates of $\alpha_{-5}, \alpha_{-4}, \dots, \alpha_{+5}$. The output of the first column of Table VII is also plotted in Figure 1. The first column shows that the yield difference is always negative, i.e. the yield is maximum on the auction date. Also, the yield difference shrinks as the

⁶This choice of clustering level is motivated by the fact that start-of-the-month auctions do not focus on the same part of the yield curve as end-of-the-month auctions. Therefore errors could be correlated at this level (See Figure 5 and Table II).

auction date approaches, and then grows back again. This means that the yield increases gradually ahead of the auction, and then reverts. Similarly, in the second column, I find that the special repo volume increases gradually ahead of the auction.⁷ Finally, in the third column, trading volume is also seen mostly increasing ahead of the auction, except for a temporary decrease at $t=-3$. Overall, the patterns found in the data are similar to the patterns predicted in Implications 1 and 2.

D. Empirical relevance of Implication 3

Empirical strategy

The empirical strategy relies on the following four components. First, I exploit the dealers' meetings and auction size announcements. I regard these events as scheduled arrival of information, as in Implication 3. That is, dealers acquire information on average during these events, either from the Treasury or from other dealers. In the robustness section, I give an indication that these events are indeed informative. In the discussion section, I further discuss the informativeness of the events.

Second, I exploit exogenous variation in the timing of information arrival. Duffie (2010) predicts that the decrease in price accentuates as the auction day approaches. To make sure not to capture the effect of Duffie (2010) and other models, one should control for the number of days that separates an observation to the auction day. This would not be possible if the information arrived at a fixed time, e.g. always four days before the auction date. Fortunately, there is an heterogeneity in the timing of the release of information. The heterogeneity is across maturities, across time periods, and in the panel. The heterogeneity is also arguably exogenous to the price pattern (See Figure 4, and Table III).

Third, I use the Spanish yield curve to control for yield movements coming from interest-rate or credit events, such as credit downgrades. I choose the Spanish yield curve because of the two following qualities. As one of the non-core countries (Greece, Ireland, Italy, Portugal and Spain), Spain's debt shares many fundamental characteristics with Italy's, and is therefore likely to be affected by the same events.⁸ Furthermore, and contrary to the Italian yield curve, the Spanish yield curve does not include any of the reopened Italian bonds. Hence, arbitrageurs are likely to find it too expensive to close the price gap between this curve and the reopened bonds, around

⁷The volume of special repo volume is an indicator of short-selling activity

⁸Table VI shows the correlation between the Italian yield curve and Spain, among other countries

auctions. The Spanish controls are therefore unlikely to wash the model's effect out.

Finally, I use dealer quotes to capture the behavior of primary dealers. Given that all primary dealers of Italian government bonds are also secondary dealers (Department of the Treasury, 2020) –and that a significant number of secondary dealers are also primary dealers⁹– using dealer quotes allows to be as close to the model as possible.

Specification

I predict that $\beta > 0$ (null hypothesis $\beta = 0$) in the following regression:

$$Yield_{n,d-t} - Yield_{n,d-t-1} = \beta * \mathbb{1}_{n,d-t}^{Info} + FE_t^{DaysToAuction} + Control_{d-t}^{InterestRate} + FE^{Time,Maturity} + \epsilon_{n,d-t} \quad (18)$$

where

$$FE_t^{DaysToAuction} = \sum_{days-to-auction \in (1,5)} \mathbb{1}_t^{days-to-auction}$$

$$Control_{d-t}^{InterestRate} = \sum_i (InterestRate_{i,d-t}^{Spain} - InterestRate_{i,d-t-1}^{Spain})$$

t is the number of days before or after the reopening date of bond n , and belongs to $(1,5)$, d denotes the reopening date, and $d-t$ is the observation date. $Yield_{n,d-t}$ measures the yield of the reopened bond n on date $d-t$. The “info dummy” $\mathbb{1}_{n,d-t}^{Info}$ takes the value 1 if $d-t$ is either the date on which the Treasury meets with dealers to discuss the reopening of bond n , or the day following the announcement of the reopening size of bond n , 0 otherwise. $FE_t^{DaysToAuction}$ consists of the sum of five dummies, each one indicating whether t is equal to 1, 2, 3, 4 or 5. $InterestRate_{i,d-t}^{Spain}$ is the Spanish benchmark interest rate of maturity i at date $d-t$, where i can be three, six, or 12 months, two to ten years, or 30 years. $FE_{n,d-t}^{Time,Maturity}$ includes time fixed effects at the bi-weekly level. It

⁹e.g. Among the 29 credit institution listed by Borsa Italiana (2020) as intermediaries in a market where Italian sovereign bonds are traded (among other bonds), 16 are also listed as primary dealer by Department of the Treasury (2020)

also includes maturity fixed effects, i.e. a dummy for each of two, three, five, seven, ten, and 30 year maturities. The standard errors are clustered at the bi-weekly level.

Results

The regression results are reported in Table VIII. The first column reports the result of a pooled specification. As predicted, I find that the coefficient of the info dummy is positive, and significantly different from zero. It is equal to 1.20 bps. This means that the yield increases by 1.20 bps more on each of these two information days, than on non-information days. The total effect is therefore of $2 \times 1.20 \text{ bps} = 2.40 \text{ bps}$. Given that the average sample duration is six years, the effect is equivalent to a circa 15 bps drop in prices. In the second column, I cluster the standard errors, I add the time and maturity fixed effects.

One way to put the magnitude of the effect into perspective is to understand how much of the pre-auction yield increase the model can explain. To that end, I turn to column 1 in Table VIII. The change in yield between $t=-6$ and $t=-1$ –without the effect of information– can be approximatively computed as the sum of the day-to-auction coefficients plus five times the constant. It is therefore of 0.72 bps. When adding the effect of information, the change in yield between $t=-6$ and $t=-1$ is equal to 3.12 bps ($0.72 + 2 \times 1.20$). The model can therefore explain 77% ($2 \times 1.20 / 3.12$) of the pre-auction increase in yield between $t=-6$ and $t=-1$.

Another way to put the magnitude of the effect into perspective is to compare it to a benchmark. One potential benchmark is the transaction cost. If the model's effect is equal to a substantial fraction of transaction costs, it would indicate that the effect is economically large. Indeed, given that bond returns are typically low, transaction costs are an important component of profitability. I restrict the sample to the period from 2004 to 2015 due to data constraints. I compute spreads from Datastream (or from MTS, when not available), expressed in yield basis points. I find that the half-spread is 0.96 bps. Then, I run the same specification as in column 2 of Table VIII, but for the period from 2004 to 2015. I find the model's effect is equal to $2 \times 1.32 \text{ bps} = 2.64 \text{ bps}$. Hence, the effect of the model's mechanism is $2.64/0.96 = 2.8$ times larger than the half-spread charged by dealers for a transaction.

Overall, as predicted by Implication 3, the yield increases on average with the scheduled release

of information regarding the net supply. The effect is statistically and economically significant.

E. Robustness

First, it is useful to verify that the two events are informative. I use an increase in trading volumes as an indication of the arrival of information.¹⁰ I test whether trading volumes increase more on event days than on other days. Table IX shows the result of a pooled and fixed-effect specification. In both columns, the coefficient of interest is positive and significant, albeit weaker in the second column. Trading volumes increase by 8 to 9% on event days than on other days.

Second, the specific dynamics during the sovereign bond crisis could be better controlled for, over and above the time fixed effects included in the baseline specification. To that end, I interact the main right-hand side dummy with a dummy equal to one for the period from 2010 to 2012, and zero otherwise. The results are shown in column 1 of Table X. I find that the coefficient of interest (the non-interacted term) is still significant.

Third, one may want to separate the pricing effect of the dealers' meeting from that of the size announcement. Column 2 in Table X shows the result of a specification that includes two dummies. I find that the coefficient of each of these dummies is significant, and is equal to 1.13bps for the dealers meeting, and 1.58bps for the size announcement.

Fourth, the results are robust to the following changes: removing from the sample the reopenings whose dates are harder to predict according to Table II's classification (see column 3 of Table X); using month-level fixed effects and clusters (see column 4 of Table X); adding ISIN or ISIN-time fixed effects; adding week-day fixed effects, or using the log clean prices instead of yields.

¹⁰The arrival of information generates trading if it increases heterogeneity among traders, e.g between an agent that possesses information and an agent that does not (Glosten and Milgrom (1985)), or between agents that disagree on the interpretation of information (Kim and Verrecchia (1994, 1997)), or between agents with different horizons (cf. this paper's model). However, it will not generate trading between identical agents that possess the same piece of information and agree on its interpretation. The presence of abnormal trading should therefore be viewed as a sufficient –and not as a necessary– indication of the arrival of information

IV. Discussion

This section discusses applications to non-Treasury markets, the external validity of the findings, as well as the fit between the theory and the two events used in the empirical section.

A. Application to Seasoned Equity Offerings (SEOs)

A SEO is an anticipated increase in the supply of a stock. One type of SEOs is known as “bought deal”, whereby the issuer states the issuance amount, then an auction is realized among investment banks, and the bank with the highest bid buys the entire issue (Gao and Ritter (2010)). The date and the issuance size are fixed, but a given investment bank might not precisely know the demand of the other banks. Therefore, the supply shock is not perfectly anticipated. Given these elements, my paper’s model can apply to this context.¹¹

Conjecture 1: *Suppose that the noise about the net supply of a stock decreases gradually as the date of a SEO approaches. Then, before the SEO, the stock price is higher than the expected offer price, and it decreases gradually. In addition, the short-selling volume is larger than usual, and increases gradually.*

Patterns similar to that in Conjecture 1 are documented in Corwin (2003), Meidan (2005), and Henry and Koski (2010). Corwin (2003) finds that there is a negative abnormal return of holding the stock five days before the SEO, and selling it one day before SEO day. Meidan (2005) finds a similar pattern. Moreover, Henry and Koski (2010) find that the mean and the median short-selling volume are abnormally high ahead of the SEO.

B. Application to predictable trades of futures contracts

Some investors roll-over their futures contracts in a predictable fashion. Bessembinder et al. (2016) study a large exchange-traded fund (ETF) which tracks oil prices, and invests in oil futures. On some predictable dates, the ETF sells its future contracts, and invests in newer contracts. The strategy is known by the market, and the trading date is announced on the ETF’s website.

¹¹ Admittedly, a SEO is not a true supply shock: the size of the SEO might send a signal about the fundamental value of the asset. Therefore, my model can apply only after controlling for the informational content of an SEO about the stock’s fundamental value.

However, the presence of buyers on the futures market at the time of the trade is unlikely to be known in advance. Therefore, the supply shock is not perfectly anticipated. Hence, my paper's model can apply to this context as well.

Conjecture 2: *Suppose that the noise about the demand for a futures contract decreases gradually as the date of a predictable sale approaches. Then, before the sale date, the price of the futures is higher than the expected sale price, and it decreases gradually.*

A price pattern similar to that in Conjecture 2 is documented in Bessembinder et al. (2016). For each day in a (-10;+10) window, they compare the price of the futures oil contract sold by the ETF ("front contract"), to the price of the contract bought by the ETF ("second contract"). They find that the price of the front contract decreases as the trade date approaches, relative to the price of the second contract.

C. External validity

In this paper, I use two important empirical features: the availability of bond prices prior to the auction, and the ability for the econometrician to observe the timing of information release. These two features are present in the US context *but* are unobservable by the econometrician. These features are present in the Italian context *and* are observable. Therefore, I chose to use Italian data. Importantly, because these features are present in both countries, the Italian results also extend to the US. That is why this paper's results can be considered as having external validity.

In detail, the first important feature used in this paper is the availability of bond prices prior to the auction. In Italy, the econometrician can precisely measure the bond price prior to the auction. This is because, in Italy, there are systematic reopenings of bonds which are already trading in the secondary market. In the US, one would have to rely on the when-issued market. However, the when-issued price is a noisy proxy because dealers act strategically in this market to infer other dealers' information (Nyborg and Sundaresan (1996)). To be clear, strategic behaviors with the intent to infer information cannot be ruled out in the Italian set-up. However, one would expect the price impact of such behaviors to be smaller in a large and liquid market such as the market studied in this paper. An alternative in the US would be to study the price of the off-the-run bond.

But one runs the risk, at best, to use a noisy proxy of the on-the-run bond price and, at worst, to capture the on/off-the-run phenomenon. The second important feature is the ability to observe the timing of the release of information. In Italy, such release occurs at the dealers' meeting. This date is perfectly observable. In the US, there is also an arrival of information. Indeed, primary dealers have the advantage of observing the orders of their customers ahead of auctions, as is in the case for dealers in many financial markets (Hortacsu and Kastl (2012)). As client orders arrive, their understanding of the net supply becomes more and more precise (Boyarchenko, Lucca and Veldkamp (2016)). However, the arrival of orders is unobservable to the econometrician.

D. Informativeness of the two events

The theory says that, if a dealer expects an arrival of information prior to the actual auction, she forms a noisy expectation about this intermediate signal. The dealer's expectation is then composed of two error terms: one that relates to the intermediate arrival of information and disappears after the arrival, and one that disappears only after the auction (This decomposition is explicit in the dynamic version of the model). The noise is therefore larger before than after the arrival of information. The dealer hedges this noise by keeping a larger (or less negative) inventory than otherwise. After the arrival, the dealer partly reverses the hedge.

In the empirics, I use the meetings between dealers and the Treasury. I also use the announcement of the auction size. In order to be eligible to test the theory, these events have to be expected by the dealers to be informative about the auction. The robustness section gives indications of that the events *are* informative. What is left to determine is whether these events *are expected to be* informative by dealers. Unless rational expectations fails, this should be the case.

V. Conclusion

I develop a theory explaining why bond prices decrease gradually in a predictable fashion ahead of Treasury auctions. I show that these price movements arise when the market perfectly anticipates the size of the sale but imperfectly the extent to which natural buyers will be present at the sale. As a result, the supply shock is imperfectly anticipated, and this situation triggers a positive hedging demand which gradually declines when the noise of the supply shock reduces. Hence, the price

decreases gradually. I illustrate the empirical relevance of the theory in the days leading up to Italian government bond issuances. I use meetings between the Treasury and primary dealers as events that reduce the supply shock noise. I find that the meetings, together with other relevant events, explain a circa 15 bps drop in Treasury prices. The findings cannot be explained by the on/off-the-run effect since that my empirical setting is free from this effect. The paper is relevant for several other instances of anticipated asset sales such as Seasoned Equity Offerings, and the rebalancing of future contracts.

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Appendix A. Figures

Figure 1. Yield pattern around auctions. Reports the result of ten t-tests. Null hypothesis: the yield t days before the auction is equal to the yield on the auction day, where t belongs to $(-5, +5) \setminus \{0\}$. The yield used in this Figure is the secondary yield of a bond which is about to be reopened. A reopening is a primary auction which results in the increase in outstanding volume of a bond that was first issued in the past. The sample is composed of 849 Italian reopenings from 2000 to 2015. The solid line is the point estimate. The two other lines correspond to the 90% interval confidence. Data from Datastream (RY datatype).

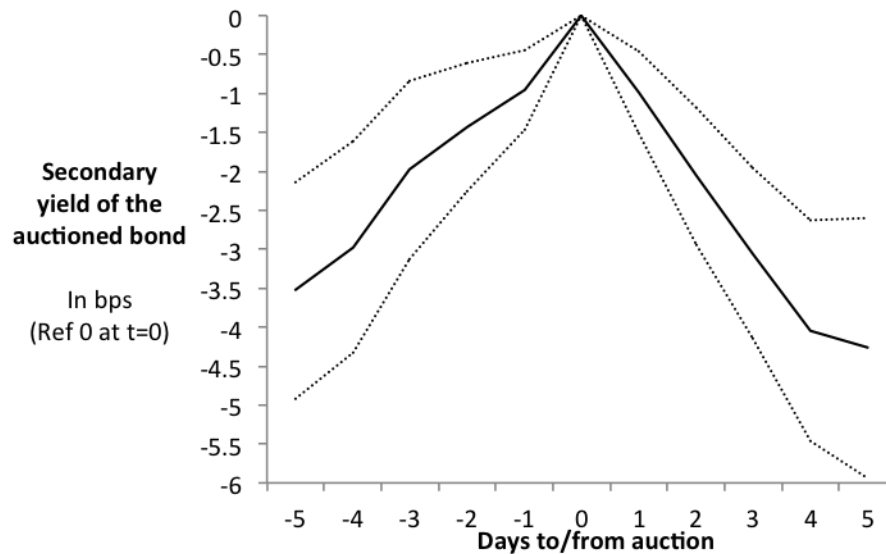


Figure 2. Model timeline

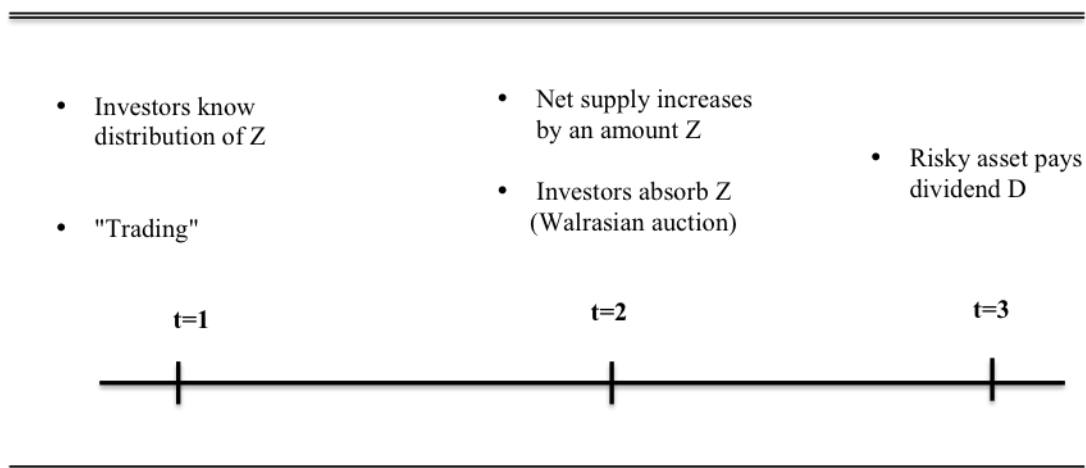
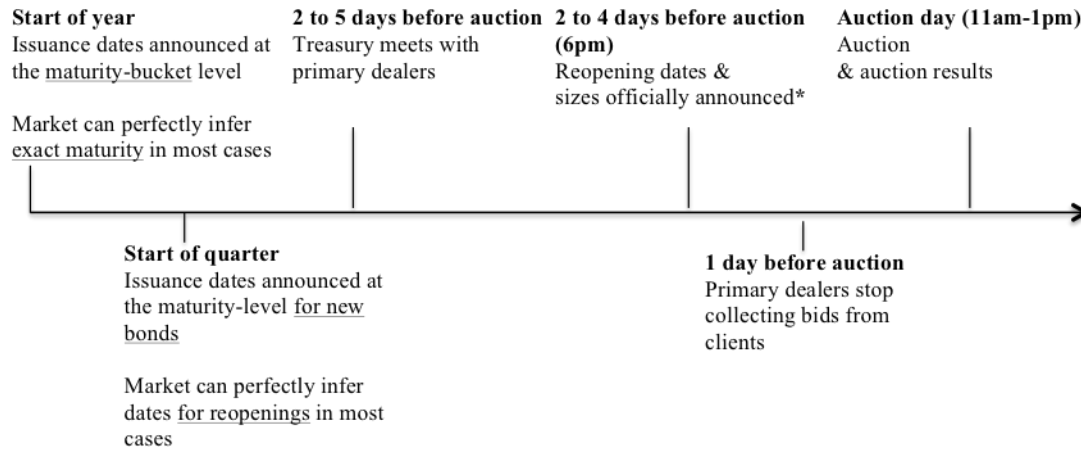


Figure 3. Italian reopenings timeline



* For 3-30Y bonds over 2000-11, the date was announced two days before the size

Figure 4. Distribution of observations across days-to-auction. For example, circa 30% of the information days occur five days before the auction

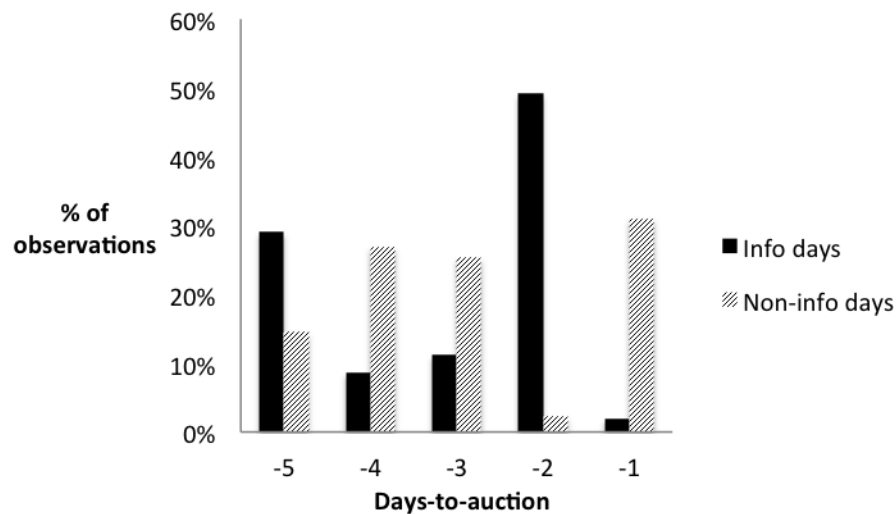
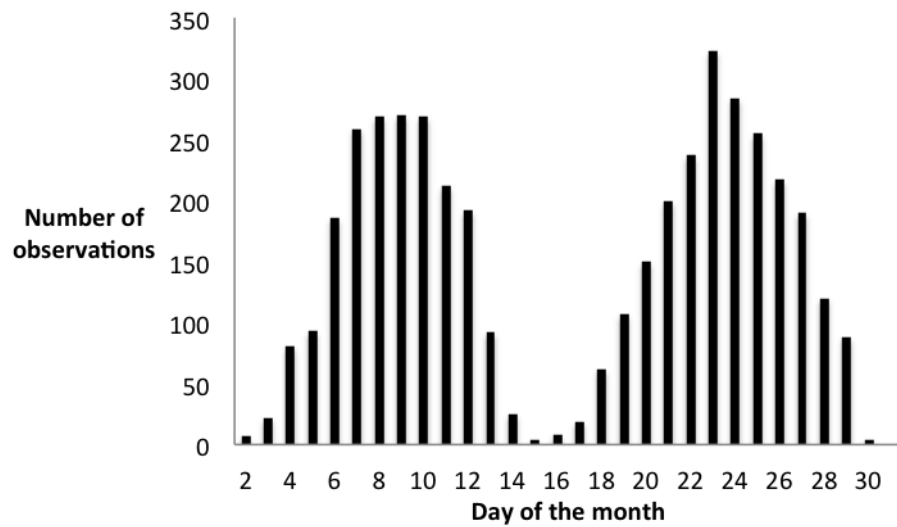


Figure 5. Distribution of observations within the month. For example, circa 250 observations occur on the 22nd of the month



Appendix B. Tables

Table I. Interpretation of the model in the context of Treasury auctions

Model	Treasury auction context
t=1	A few days prior to the auction
t=2	Auction date
t=3	Re-sale or maturity of the bond
Investors	Primary dealers
Implicit seller at t=2	The Treasury, after catering to Natural Buyers (foreign, investment funds, individuals)
$\bar{\theta}$	Initial inventory size of the dealers, > 0 or < 0
Z	Change in inventory due to issuance = Difference between issue size and demand of Natural Buyers
$\bar{Z}$	Expected change in inventory = What Primary Dealers expect to buy at auction (> 0)
$\bar{Z} - Z$	Unexpected demand of Natural Buyers
If $Z > 0$	Dealers increase inventory: they provide liquidity to (=buy from) Treasury
If $Z < 0$	Dealers decrease inventory: they provide liquidity to (=sell to) Natural Buyers
If $Z > \bar{Z}$ or $Z < -\bar{Z}$	Dealers provide more liquidity than expected
If $-\bar{Z} < Z < \bar{Z}$	Dealers provide less liquidity than expected

Table II. Historical reopening frequency of Italian sovereign bonds from 2000 to 2015. Bonds can be reopened in the first half (“start”), or the second half (“end”) of the month. For example, five-year bonds are issued once a month at the start of the month during 2000-11, and at the end of the month during 2012-15. A potential exception is defined as a calendar month when there was no reopening on that month for at least two of the 16 years of the sample. A reopening is a primary auction which results in the increase in the outstanding volume of a bond which was first issued in the past.

Maturity	Reopening frequency	Part of month	Potential exceptions
30Y	Unclear	Start	
15Y	Unclear	Start	
10Y	1/month	End	Nov
7Y	1/month (2013-15)	Start	Aug, Dec
5Y	1/month (2000-15)	Start (2000-11), End (12-15)	Aug, Nov, Dec
3Y	1/month (2004-15)	End (2004-11), Start (12-15)	Aug, Nov, Dec
	2/month (2000-03)		
2Y	1/month (2000; 02-15)	End	Nov, Dec
	2/month (2001-02)		

Table III. Dates of dealers’ meeting (D) and auction size announcement (S) relatively to the auction date. For example, for five and ten-year bonds from 2000 to 2011, the dealers’ meeting takes place five business days before the reopening date. For the auction size announcement date, the next business day is reported because the announcement is made after market close. There are some exceptions to these dates. The most frequent exceptions are reported in parenthesis, provided they occur at or after t-5. Most exceptions before 2012 come from the fact that, for some months, the Treasury introduced a one-day difference (instead of the typical zero-day difference) between the announcement of the auction size of two-year auctions and the announcement of the auction date of long-term bonds. Most exceptions after 2012 come from the postponement or advancement of announcements which would otherwise have occurred during official Italian holidays which are not market holidays. These exceptions – as well as the more persistent changes of 2012 and during 2013-2015 – are known at latest at the start of the year.

Maturity	2000-11		2012		2013-15	
	D	S	D	S	D	S
2Y	-3 (-4)	-2	-2	-1	-3 (-2)	-2 (-1)
5/10Y	-5	-2	-4 (-5, -3)	-2 (-3, -1)	-5 (-3)	-2 (-3)
3/7/15/30Y	-5	-2	-4 (-5, -3)	-2 (-3)	-4 (-3)	-2

Table IV. Sample summary statistics - Italian Treasury reopenings (2000-15). An on-the-run bond is defined as the most recent bond for a given maturity. The remaining maturity is the maturity of the bond at the time of the reopening. The reissued amount is the size of the reopening. The bid-cover ratio is defined as the ratio of the bid volume to the offered volume. A reopening is a primary auction which results in the increase in the outstanding volume of a bond which was first issued in the past.

Maturity	On/off run	Obs.	Remaining maturity (Years)		Reissued amount (€MM)		Bid-cover ratio	
			Mean	Std.	Mean	Std.	Mean	Std.
30Y	On	67	29.73	2.63	1,514	647	1.86	1.57
	Off	11	24.97	2.73	952	406	1.89	0.4
15Y	On	53	14.78	1.68	1,819	547	1.63	0.31
	Off	27	11.77	2.30	1,256	605	1.74	0.31
10Y	On	140	10.09	0.22	2,645	584	1.57	0.34
	Off	44	8.22	1.84	1,604	786	1.71	0.31
7Y	On	17	7.06	0.18	2,367	282	1.50	0.11
5Y	On	144	4.84	0.37	2,393	683	1.73	0.43
	Off	8	3.69	0.33	730	205	2.28	0.48
3Y	On	172	2.82	0.15	2,407	745	1.79	0.48
	Off	3	2.74	0.24	1,760	885	1.88	0.71
2Y	On	157	1.81	0.17	2,069	615	2.09	0.68
	Off	6	1.80	0.16	1,917	376	1.88	0.08

Table V. Sample summary statistics – Secondary and Repo market variables for reopened bonds. The five-day pre-auction yield change is defined as the difference between the yield at $t=0$ and the yield at $t=-5$, where $t=0$ denotes the reopening date. Columns 3 and 4 report the mean and the standard deviation of this measure. Columns 5 and 6 (7 and 8) report the mean and the standard deviation of bond trading (special Repo trading volume) for 11 days around the reopening dates. Yields and trading volumes are from Datastream. Repo trading volumes data are from the MTS Repo platform, and are available over a subsample (from 2005 to October 2012). A special Repo contract is a cash loan agreement where the ISIN of the bond serving as collateral is explicitly designated. An on-the-run bond is defined as the most recent bond for a given maturity. A reopening is a primary auction which results in the increase in the outstanding volume of a bond which was first issued in the past.

Maturity	On/off run	Five-day Δ yield (bps)		Daily trading volume (€Billion)		Special repo vol. (€MM)	
		Mean	Std.	Mean	Std.	Mean	Std.
30Y	On	4.84	11.26	16.28	42.73	556.51	362.25
	Off	4.57	12.13	55.38	105.61	458.71	274.39
15Y	On	3.43	12.97	19.47	32.76	669.53	366.10
	Off	12.44	19.60	9.72	20.98	548.13	342.58
10Y	On	2.58	10.76	13.57	30.49	1,006.89	540.78
	Off	12.68	24.63	2.91	12.13	858.65	584.42
7Y	On	-2.72	8.53	26.38	24.64	na	na
5Y	On	2.31	15.78	6.67	15.54	760.58	521.12
	Off	5.42	33.46	3.43	6.57	485.65	238.41
3Y	On	2.43	15.58	1.68	5.41	655.77	383.98
	Off	-27.41	39.05	.01	.01	429.89	221.52
2Y	On	3.35	17.21	1.98	5.87	533.89	435.84
	Off	-1.87	10.96	.00	.00	864.26	435.47

Table VI. Correlation between the Italian and the yield curves of several other European countries from 2000 to 2015. For example, there is a correlation of 0.96 between ten-year Italian bonds and ten-year Spanish bonds. The data come from Bloomberg (I40, I61, I84, I62, I14 and I16 curves). The numbers of observations are reported in parenthesis.

Maturity	Corr. (Spain, Italy)	Corr. (Portugal, Italy)	Corr. (Ireland, Italy)	Corr. (France, Italy)	Corr. (Germany, Italy)
3M	0.98 (4,001)	0.91 (2,887)	0.94 (3,904)	0.94 (4,040)	0.94 (4,090)
6M	0.99 (4,020)	0.90 (3,041)	0.91 (3,321)	0.91 (4,051)	0.90 (4,106)
1Y	0.98 (4,020)	0.73 (3,480)	0.72 (3,306)	0.87 (4,007)	0.85 (4,113)
2Y	0.97 (4,170)	0.59 (4,168)	0.67 (3,061)	0.83 (4,170)	0.79 (4,167)
3Y	0.96 (4,173)	0.58 (4,158)	0.67 (2,877)	0.78 (4,174)	0.72 (4,173)
4Y	0.96 (4,165)	0.60 (4,100)	0.82 (2,089)	0.76 (4,172)	0.69 (4,174)
5Y	0.96 (4,173)	0.61 (4,113)	0.75 (3,459)	0.73 (4,173)	0.64 (4,172)
6Y	0.95 (4,172)	0.61 (3,991)	0.83 (1,840)	0.73 (4,174)	0.64 (4,172)
7Y	0.95 (4,173)	0.60 (4,005)	0.96 (1,681)	0.72 (4,174)	0.62 (4,174)
8Y	0.95 (4,174)	0.62 (3,548)	0.76 (1,753)	0.73 (4,174)	0.62 (4,173)
9Y	0.95 (4,094)	0.64 (3,957)	0.79 (1,985)	0.73 (4,173)	0.61 (4,173)
10Y	0.96 (4,174)	0.68 (3,989)	0.72 (3,642)	0.70 (4,171)	0.57 (4,171)
30Y	0.94 (4,170)	0.80 (2,284)	0.93 (236)	0.68 (4,172)	0.52 (4,169)

Table VII. Table for Implications 1 and 2. I check that the patterns present in the data are compatible with Implications 1 and 2. The table reports the coefficients of t-test specifications which test the null hypothesis for the difference in yield (or in log trading volume, or log special repo volume) between date t and the reopening day, where t belongs to (-5,+5), and 0 denotes the reopening day. The specifications include fixed effects and clustering at the year-month-half ("bi-weekly") level: each month of each year is separated into a first and a second half. Sample: all two to 30-year fixed-rate Italian sovereign bonds reopened from 2000 to 2015 for which yield data are available on Datastream. A reopening is a primary auction that results in the increase in the outstanding volume of a bond that was first issued in the past. t statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

	Δ Yield (bps)	Δ Special repo log diff.	Δ Trading log diff.
t=-5 vs. t=0	-3.53*** (-4.16)	-31.36*** (-11.75)	-55.22*** (-11.82)
t=-4 vs. t=0	-2.97*** (-3.61)	-27.94*** (-10.42)	-52.68*** (-11.76)
t=-3 vs. t=0	-1.98*** (-2.85)	-28.43*** (-10.18)	-58.38*** (-18.06)
t=-2 vs. t=0	-1.43*** (-2.85)	-23.66*** (-10.96)	-50.81*** (-14.78)
t=-1 vs. t=0	-0.95*** (-3.06)	-15.75*** (-7.21)	-41.72*** (-16.19)
t=+1 vs. t=0	-0.99*** (-3.10)	-45.71*** (-11.77)	-27.04*** (-10.33)
t=+2 vs. t=0	-2.06*** (-3.85)	-55.63*** (-13.27)	-38.60*** (-11.80)
t=+3 vs. t=0	-3.06*** (-4.60)	-55.60*** (-13.30)	-38.36*** (-12.29)
t=+4 vs. t=0	-4.05*** (-4.68)	-56.54*** (-13.72)	-39.73*** (-12.41)
t=+5 vs. t=0	-4.27*** (-4.19)	-57.06*** (-13.18)	-40.30*** (-12.87)
Sample period	2000-15	2004-12	2000-15
Cluster level	Bi-weekly	Bi-weekly	Bi-weekly
Auction sample	849	414	668

Table VIII. Main table. I test if the yield increases more after the arrival of information than on non-information days, as predicted in Implication 3 in the context of a large sample. The left-hand side variable is the one-day change in the yield of the reopened bond between t and $t-1$, where t belongs to a $(-5, -1)$ window before the reopening date, and 0 is the reopening date. The variable of interest is 1^{Info} which takes the value 1 if t is either the day on which the Treasury meets with dealers, or the day following the announcement of the auction size, 0 otherwise. The interest-rate controls consist of daily changes in benchmark interest rates of the Spanish yield curve. The time fixed effects and clustering are at the year-month-half (“bi-weekly”) level: each month of each year is divided into a first and a second half. The days-to-auction fixed effects consist of the sum of five dummies, each one indicating whether t is equal to -1 , -2 , -3 , -4 or -5 . The dummy corresponding to $t=-5$ is used as a basis, and thus set to zero. The maturity fixed effects correspond to a dummy for each of two, three, five, seven, ten, and 30 year maturities. The constant and the coefficients of the days-to-auction dummies are reported for the pooled specification. The left-hand side yield data comes from Datastream (RY datatype). The Spanish yield curve data come from Bloomberg (I67 Bid curve). Sample: two to 30-year fixed-rate Italian sovereign bonds reopened from 2000 to 2015. A reopening is a primary auction that results in the increase in the outstanding volume of a bond that was first issued in the past. t statistics in parenthesis. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$

	Δ Yield One-day change (bps) <i>Pooled spec.</i>	Δ Yield One-day change (bps) <i>FE spec.</i>
1^{Info} (Variable of interest)	1.201*** (5.84)	1.198*** (3.72)
$1_{t=-1}$	1.311*** (5.11)	
$1_{t=-2}$	0.226 (0.91)	
$1_{t=-3}$	1.330*** (5.42)	
$1_{t=-4}$	1.249*** (5.05)	
Constant	-0.679*** (-3.39)	
Interest-rate controls	Yes	Yes
Days-to-auction fixed effect	Yes (Coeff. reported)	Yes
Time FE and cluster	None	Bi-weekly
Other fixed effects	None	Maturity
Observations	3,985	3,984
Adj. R^2	0.57	0.60

Table IX. Informativeness table. I test if the trading volume increases more after the arrival of information than on non-information days. The left-hand side variable is the one-day log change in the trading volume of the reopened bond between t and $t-1$, where t belongs to a $(-5, -1)$ window before the reopening date, and 0 is the reopening date. The variable of interest is 1^{Info} which takes the value 1 if t is either the day on which the Treasury meets with dealers, or the day following the announcement of the auction size, 0 otherwise. Includes time fixed effects and clustering at the year-month-half ("bi-weekly") level: each month of each year is divided into a first and a second half. The days-to-auction fixed effects consist of the sum of five dummies, each one indicating whether t is equal to -1, -2, -3, -4 or -5. The maturity fixed effects correspond to a dummy for each of two, three, five, seven, ten, and 30 year maturities. The left-hand side yield data comes from Datastream (RY datatype). The trading volume data comes from Datastream (VA datatype). Sample: two to 30 year fixed-rate Italian sovereign bonds reopened from 2000 to 2015. A reopening is a primary auction that results in the increase in the outstanding volume of a bond that was first issued in the past. t statistics in parenthesis. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$

	Δ Trading vol One-day change (log diff) <i>Pooled spec.</i>	Δ Trading vol One-day change (log diff) <i>FE spec.</i>
1^{Info}	0.081** (2.33)	0.089* (1.67)
Days-to-auction fixed effect	Yes	Yes
Time FE and cluster	No	Bi-weekly
Other fixed effects	No	Maturity
Observations	3,334	3,334
Adj. R^2	0.01	0.00

Table X. Robustness table. I test if the yield increases more after the arrival of information than on non-information days, as predicted in Implication 3 in the context of a large sample. The left-hand side variable is the one-day change in the yield of the reopened bond between t and $t-1$, where t belongs to a $(-5, -1)$ window before the reopening date, and 0 is the reopening date. The variable of interest is 1^{Info} which takes the value 1 if t is either the day on which the Treasury meets with dealers, or the day following the announcement of the auction size, 0 otherwise. In column 1, I control for the specific dynamics of the sovereign bond crisis by adding a dummy equal to 1 on 2010-2012, 0 otherwise. Note that the coefficient corresponding to the dummy is absorbed by the time fixed effects. In column 2, I separate the effect of the dealer meeting from the effect of the size announcement. I create two dummies. The dealer (resp. size) dummy takes the value 1 if t is the day on which the Treasury meets with dealers (resp. the day following the announcement of the auction size), 0 otherwise. In column 3, I keep only "predictable" reopenings, defined as reopenings which dates can be perfectly predicted by the econometrician before the corresponding dealers' meeting as indicated in appendix Table II. Specifically, are removed from the sample reopenings which frequency is labelled as "unclear" as well as reopenings labelled as "exceptions" in Table II. Column 1 - 3 include time fixed effects and clustering at the year-month-half ("bi-weekly") level: each month of each year is divided into a first and a second half. Column 4 includes fixed effects and clustering at the year-month ("monthly") level. The interest-rate controls consist in daily changes in benchmark interest rates of the Spanish yield curve. The days-to-auction fixed effects consist of the sum of five dummies, each one indicating whether t is equal to -1, -2, -3, -4 or -5. The dummy corresponding to $t=-5$ is used as a basis, and thus set to zero. The maturity fixed effects correspond to a dummy for each of two, three, five, seven, ten, and 30 year maturities. The left-hand side yield data comes from Datastream (RY datatype). The Spanish yield curve data come from Bloomberg (I67 Bid curve). Sample: two to 30 year fixed-rate Italian sovereign bonds reopened from 2000 to 2015. A reopening is a primary auction that results in the increase in the outstanding volume of a bond that was first issued in the past. t statistics in parenthesis. $*p < 0.10$, $**p < 0.05$, $***p < 0.01$

	Δ Yield One-day change <i>Control for crisis</i>	Δ Yield One-day change <i>Two betas</i>	Δ Yield One-day change <i>Predictable</i>	Δ Yield One-day change <i>Monthly FE</i>
1^{Info}	1.125*** (3.95)		0.922*** (2.99)	1.192*** (3.63)
$1^{Info} \times 1^{Crisis}$	0.277 (0.42)			
$1^{DealerMeeting}$		1.131*** (3.06)		
$1^{SizeAnnounce}$		1.579*** (3.55)		
Interest-rate controls	Yes	Yes	Yes	Yes
Days-to-auction fixed effect	Yes	Yes	Yes	Yes
Time FE and cluster	Bi-weekly	Bi-weekly	Bi-weekly	Monthly
Other fixed effects	Maturity	Maturity	Maturity	Maturity
Observations	3,984	3,984	2,476	3,984
Adj. R^2	0.60	0.60	0.65	0.60

Appendix C. Proofs

Proof of Lemma 1:

$$V_2(Z, W_2) = -\exp - \left\{ \alpha \left(W_2(Z) + \frac{1}{2} \alpha \sigma^2 (\bar{\theta} + Z)^2 \right) \right\} \quad (C1)$$

where $W_2(Z) = C_0 - (\theta_1 - \theta_0)P_1 + \theta_1(\bar{D} - \alpha\sigma^2(\bar{\theta} + Z))$

For all Z , $V_2(Z_1 + Z) = V_2(Z_1 - Z)$ where $Z_1 = \theta_1 - \bar{\theta}$. Therefore, $V_2(Z)$ is symmetric in Z_1 .

Furthermore, $\frac{\partial V_2(Z)}{\partial Z}$ is positive when $Z > Z_1$. Therefore, $V_2(Z)$ is an increasing function of Z over $[Z_1; +\infty)$. Therefore, due to the symmetry, $V_2(Z)$ is decreasing over $(-\infty; Z_1]$.

Finally, $\frac{\partial^2 V_2(Z)}{\partial^2 Z}$ is negative when $Z > Z_1 + \frac{1}{\alpha\sigma}$. Therefore, $V_2(Z)$ is a concave function of Z over $[Z_1 + \frac{1}{\alpha\sigma}; +\infty)$.

■

Proof of Proposition 1:

The squared term in Z in equation 8 is not normally distributed. However, using lemma 1 in Vayanos and Wang (2012), the problem can be reduced to a mean-variance problem. Specifically, I first find that investors' objective function is:

$$\mathbb{E}_Z \left[-\exp \left\{ -\alpha \left(W_1 + \theta_1 (\mathbb{E}_Z(P_2^*) - P_1) + \frac{\alpha}{2} \sigma^2 (\bar{\theta} + \bar{Z})^2 + (Z - \bar{Z}) \left(-\alpha \sigma^2 \theta_1 + \alpha \sigma^2 (\bar{\theta} + \bar{Z}) \right) + (Z - \bar{Z})^2 \frac{\alpha}{2} \sigma^2 \right) \right\} \right] \quad (C2)$$

Using Vayanos and Wang (2012)'s notation, investors' objective function can be written as:

$$\mathbb{E}_Z \left[-\exp \left\{ -\alpha \left(A + (Z - \bar{Z})B + (Z - \bar{Z})^2 \frac{C}{2} \right) \right\} \right] \quad (C3)$$

With $A = W_1 + \theta_1 (\mathbb{E}_Z(P_2^*) - P_1) + \frac{\alpha}{2} \sigma^2 (\bar{\theta} + \bar{Z})^2$; $B = -\alpha \sigma^2 \theta_1 + \alpha \sigma^2 (\bar{\theta} + \bar{Z})$; $C = \alpha \sigma^2$

Noting that $(Z - \bar{Z}) \sim N(0; \sigma_Z^2)$, I can apply lemma 1 in Vayanos and Wang (2012). More precisely, the problem is equivalent to

$$\text{Max}_{\theta_1} \mathbb{E}_Z \left[-\exp \left\{ -\alpha \left(A + B(Z - \bar{Z}) \right) \right\} \right] \quad (C4)$$

and replacing σ_Z^2 with $\sigma_Z^2(1 + \alpha C \sigma_Z^2)^{-1}$

We now have a mean-variance problem. Investors' problem is equivalent to:

$$Max_{\theta_1} \left(\mathbb{E}_Z \left(A + B(Z - \bar{Z}) \right) - \frac{1}{2} \alpha Var \left(A + B(Z - \bar{Z}) \right) \right) \quad (C5)$$

i.e. equivalent to

$$Max_{\theta_1} \left[\theta_1 \left(\mathbb{E}_Z(P_2) - P_1 \right) - \frac{\alpha}{2} \left(\theta_1^2 \frac{Var_Z(P_2)}{1 + \alpha^2 \sigma^2 \sigma_Z^2} + 2\theta_1 \alpha \sigma^2 (\bar{\theta} + \bar{Z}) \frac{Cov(P_2, Z)}{1 + \alpha^2 \sigma^2 \sigma_Z^2} \right) \right] \quad (C6)$$

One then obtains the optimal demand function by using the first order conditions.

■

Proof of Proposition 2:

$$\mathbb{E}_Z(P_2) - P_1^* = -\frac{\sigma^4 \sigma_Z^2 \alpha^3 \bar{Z}}{1 + \alpha^2 \sigma^2 \sigma_Z^2} < 0 \quad (C7)$$

$$\frac{\partial(\mathbb{E}_Z(P_2) - P_1^*)}{\partial \sigma_Z^2} = -\frac{\sigma^4 \alpha^3 \bar{Z}}{(1 + \alpha^2 \sigma^2 \sigma_Z^2)^2} < 0 \quad (C8)$$

■

Proof of Lemma 2

In an economy where investors care only about one-period returns, CARA short-sighted investors maximize at $t=1$

$$\mathbb{E}_Z \left[-\exp \left\{ -\alpha (\theta_1 P_2 + C_0 - (\theta_1 - \theta_0) P_1) \right\} \right] \quad (C9)$$

i.e. the expectation over net supply Z of the value $\theta_1 P_2$ of the total risky portfolio at $t=2$, plus the endowment in cash C_0 minus the cost $\theta_1 - \theta_0 P_1$ of the additional risky position taken at $t=1$.

This is a mean-variance problem. The equilibrium price is equal to

$$P_1^{*,shortsighted} = \mathbb{E}_Z(P_2) - \alpha Var_Z(P_2) \bar{\theta} \quad (C10)$$

Therefore $P_1^{*,shortsighted} < P_1^*$.

In addition, in an economy where investors do not expect any sale, the equilibrium price is given by setting Z equal to zero in (5). More precisely, it is equal to

$$P_1^{*,nosale} = P_2^{*,nosale} = \bar{D} - \alpha \sigma^2 \bar{\theta} \quad (C11)$$

I find that $P_1^* - P_1^{*,nosale} = -\frac{\alpha\sigma^2}{1+\alpha^2\sigma^2\sigma_Z^2}\bar{Z} < 0$

■

Proof of Proposition 3 (Extension 1)

I find the following equilibrium prices:

$$P_{2.5}^{*,Extension1} = \bar{D} - \alpha\sigma^2(\bar{\theta} + Z - Z_{2.5}) \quad (C12)$$

$$P_2^{*,Extension1} = \mathbb{E}_{Z_{2.5}}(P_{2.5}^{*,Extension1}) - \alpha^3\sigma^4\kappa^2\bar{Z} \quad (C13)$$

$$P_1^{*,Extension1} = \mathbb{E}_Z(P_2^{*,Extension1}) + \alpha^5\sigma^6\kappa^4\bar{Z} \quad (C14)$$

The proof of the equilibrium price at t=2.5 is standard. That of the equilibrium price at t=2 is identical to that of Proposition 1, except one uses $-Z_{2.5}$ instead of Z , and $\bar{\theta} + Z$ instead of $\bar{\theta}$.

The proof of the equilibrium price at t=1 is the following. I first define the investors' value function at t=2. To do so, I take Equation C5 and use $Z_{2.5} - \bar{Z}$ instead of $Z - \bar{Z}$, and $\bar{\theta} + Z$ instead of $\bar{\theta}$.

The value function at t=2 is therefore (up to a scalar factor):

$$-exp\left\{-\alpha\left(\mathbb{E}_{Z_{2.5}}\left(A_2^* + B_2^*(Z_{2.5} - \bar{Z})\right) - \frac{1}{2}\alpha Var_{Z_{2.5}}\left(A_2^* + B_2^*(Z_{2.5} - \bar{Z})\right)\right)\right\} \quad (C15)$$

With $A_2^* = W_2 + \theta_2^*\left(\mathbb{E}_{Z_{2.5}}(P_{2.5}^*) - P_2^*\right) + \frac{\alpha}{2}\sigma^2(\bar{Z} - \bar{\theta} - Z)^2$ $B_2^* = \alpha\sigma^2(\bar{Z} - \bar{\theta} - Z + \theta_2^*)$, and replacing σ_Z^2 with κ^2 .

The investors' problem at t=1 consists in maximizing the expectation over Z_1 of the value function given in Equation (C15). After some algebra, I find that investors' objective function at t=1 to be:

$$\mathbb{E}_Z\left[-exp\left\{-\alpha\left(A_1 + (Z - \bar{Z})B_1 + (Z - \bar{Z})^2\frac{C_1}{2}\right)\right\}\right] \quad (C16)$$

With $A_1 = W_1 + \theta_1\left(\mathbb{E}_{Z_1}(P_2^*) - P_1^*\right) + \frac{\alpha}{2}\sigma^2(\bar{\theta} + \bar{Z})^2 - \alpha\sigma^2(1 - \alpha^2\sigma^2\kappa^2)\bar{Z}(\bar{\theta} + \bar{Z}/2)$; $B_1 = \alpha\sigma^2(\alpha^2\sigma^2\kappa^2\bar{Z} - \theta_1 + \bar{\theta})$; $C_1 = \alpha\sigma^2$

Noting that $(Z - \bar{Z}) \sim N(0; \sigma_Z^2)$, one can apply lemma 1 in Vayanos and Wang (2012). More precisely, the problem is equivalent to:

$$Max_{\theta_1}\mathbb{E}_Z\left[-exp\left\{-\alpha\left(A_1 + B_1(Z - \bar{Z})\right)\right\}\right] \quad (C17)$$

and replacing σ_Z^2 with κ^2

We now have a mean-variance problem. One can then obtain the optimal demand function by using the first order conditions.

■

Proof of Proposition 4 (Extension 2)

For brevity, only the equilibrium at $t=1$ is given. When needed, I use the upper-script *Extension2*.

The equilibrium price at $t=1$, and the holding of investors B are:

$$P_1^{*,Extension2} = \mathbb{E}_Z(P_2^{Extension2}) + \frac{\alpha^3 \sigma^4 \sigma_Z^2 \bar{Z}}{1 + \delta + \alpha^2 \sigma^2 \sigma_Z^2} \quad (C18)$$

$$\theta_{1,B}^* = \bar{\theta} + \frac{\delta \bar{Z}}{1 + \delta + \alpha^2 \sigma^2 \sigma_Z^2} \quad (C19)$$

Note that

$$\theta_{1,A}^* = \frac{-\bar{Z}}{1 + \delta + \alpha^2 \sigma^2 \sigma_Z^2} < 0 \quad (C20)$$

which means that investor A is short-selling at $t=1$.

Also, we have:

$$\frac{\partial \delta \theta_{1,A}^*}{\partial \sigma_Z^2} = \frac{\delta \bar{Z} \alpha^2 \sigma^2}{(1 + \delta + \alpha^2 \sigma^2 \sigma_Z^2)^2} > 0 \quad (C21)$$

■

Proof of Implication 1

Let us compare the equilibrium prices at $t=1$ when σ_Z^2 changes to $\sigma_Z^2 + \Delta_1$ and when it changes to $\sigma_Z^2 + \Delta_2$, where $\Delta_2 < \Delta_1 < 0$. P_1^* , P_1^{*,Δ_1} and P_1^{*,Δ_2} denote the three respective prices.

Given that $\frac{\partial P_1^*}{\partial \sigma_Z^2} > 0$ as indicated in (C8), we have $P_1^* > P_1^{*,\Delta_1} > P_1^{*,\Delta_2}$.

■

Proof of Implication 2

Let us compare the holdings of investors A at $t=1$ in the extension of the model, when σ_Z^2 changes to $\sigma_Z^2 + \Delta_1$ and when it changes to $\sigma_Z^2 + \Delta_2$, where $\Delta_2 < \Delta_1 < 0$. $\theta_{1,A}^*$, $\theta_{1,A}^{*,\Delta_1}$ and $\theta_{1,A}^{*,\Delta_2}$ denote the three respective holdings.

Given that $\frac{\partial \theta_{1,A}^*}{\partial \sigma_Z^2} > 0$ as indicated in (C21), we have $0 > \theta_{1,A}^* > \theta_{1,A}^{*,\Delta_1} > \theta_{1,A}^{*,\Delta_2}$.

■

Proof of Implication 3

Scheduled pieces of information are defined as reducing the noise of the net supply, as well as changing the expectation of the net supply. Let us then change $\bar{Z}$ by an amount ϵ , where $\epsilon > 0$

(i.e. the net supply is larger than expected, so the piece of information is negative); or $\epsilon < 0$ (i.e. the net supply is smaller than expected, so the piece of information is positive), and $\bar{Z} + \epsilon > 0$. Let us also change σ_Z^2 by an amount Δ where $\Delta < 0$, and $\sigma_Z^2 + \Delta > 0$. P_1^* and $P_1^{*,New}$ denote the price without and with the changes, respectively. Let us study by how much the equilibrium price changes at $t=1$ in absolute value, depending on the sign of ϵ .

Using (11) and (5), we have

$$\left| P_1^{*,New} - P_1^* \right| = \left| \left(\bar{D} - \alpha\sigma^2(\bar{\theta} + \bar{Z} + \epsilon) + \frac{\alpha^3\sigma^4(\sigma_Z^2 + \Delta)(\bar{Z} + \epsilon)}{1 + \alpha^2\sigma^2(\sigma_Z^2 + \Delta)} \right) - \left(\bar{D} - \alpha\sigma^2(\bar{\theta} + \bar{Z}) + \frac{\alpha^3\sigma^4\sigma_Z^2\bar{Z}}{1 + \alpha^2\sigma^2\sigma_Z^2} \right) \right| \quad (C22)$$

It simplifies as

$$\left| P_1^{*,New} - P_1^* \right| = \left| \frac{\alpha^3\sigma^4\Delta\bar{Z} - \alpha\sigma^2(1 + \alpha^2\sigma^2\sigma_Z^2)\epsilon}{(1 + \alpha^2\sigma^2(\sigma_Z^2 + \Delta))(1 + \alpha^2\sigma^2\sigma_Z^2)} \right| \quad (C23)$$

The denominator is positive, $\alpha^3\sigma^4\Delta\bar{Z}$ is negative, and $\alpha\sigma^2(1 + \alpha^2\sigma^2\sigma_Z^2)\epsilon$ is of the sign of ϵ . Hence, for any value of $|\epsilon|$, $\left| P_1^{*,New} - P_1^* \right|$ is larger when $\alpha\sigma^2(1 + \alpha^2\sigma^2\sigma_Z^2)\epsilon$ is positive, i.e. when $\epsilon > 0$ (negative information). One consequence is that the average change is negative, when there are as many positive as negative ϵ .

■

Proof of Implication 4

When the auction size varies with the natural demand, the net supply is more precisely determined. Hence, the proof of Implication 1 also proves Implication 4.

■

Dynamic proofs of Implications 1 - 3

The empirical implications have been presented as being generated by comparative statics. Here, I show that the implications go through in a dynamic context. For that, I derive the equilibrium prices in a model featuring an intermediate date ($t=1.5$) where σ_Z^2 is scheduled to decrease. The goal is to show that the price after the decrease in noise is, on average, lower than the price before. At $t=2$, the net supply of the risky asset is increased by $Z = Z_2 + Z_{1.5}$. Z_2 is certain at $t=2$ with a known distribution of $Z \sim N(\bar{Z}_2; \sigma_{Z_2}^2)$ in the previous dates. At $t=1.5$, $Z_{1.5}$ is revealed, with a known distribution of $Z \sim N(\bar{Z}_{1.5}; \sigma_{Z_{1.5}}^2)$ in the previous date. It is assumed that $\bar{Z}_2 = \bar{Z}_{1.5} = \bar{Z}/2$, that $\alpha^2\sigma^2\kappa_2^2 < 1$ where $\kappa_2^2 = \frac{\sigma_{Z_2}^2}{1 + \alpha^2\sigma^2\sigma_{Z_2}^2}$, and that Z_2 and $Z_{1.5}$ are independent variables. The rest of the setting is identical to the baseline model.

In the proof of Proposition 1, I replace σ_Z^2 by $\sigma_{Z_2}^2$, and $\bar{Z}$ by $Z_{1.5} + \bar{Z}/2$. I then obtain the equilibrium price at $t=1.5$:

$$P_{1.5}^* = \mathbb{E}_{Z_2}(P_2^*) + \alpha^3 \sigma^4 \kappa_2^2 (Z_{1.5} + \bar{Z}/2) \quad (\text{C24})$$

To derive the equilibrium price at $t=1$, I first define the investors' value function at $t=1.5$. To do so, I take equation (C5) and use $Z_2 - \bar{Z}/2$ instead of $Z - \bar{Z}$. The value function at $t=1.5$ is therefore the following (up to a scalar factor):

$$-exp \left\{ -\alpha \left(\mathbb{E}_{Z_2} \left(A_{1.5}^* + B_{1.5}^* (Z_2 - \bar{Z}/2) \right) - \frac{1}{2} \alpha Var \left(A_{1.5}^* + B_{1.5}^* (Z_2 - \bar{Z}/2) \right) \right) \right\} \quad (\text{C25})$$

With $A_{1.5}^* = W_{1.5} + \theta_{1.5}^* \left(\mathbb{E}_{Z_{1.5}}(P_2^*) - P_{1.5}^* \right) + \frac{\alpha}{2} \sigma^2 (\bar{\theta} + (Z_{1.5} + \bar{Z}/2))^2$; $B_{1.5}^* = \alpha \sigma^2 (\bar{\theta} - \theta_{1.5}^* + Z_{1.5} + \bar{Z}/2)$ and replacing $\sigma_{Z_2}^2$ with κ_2^2

The investors' problem at $t=1$ consists in maximizing the expectation over $Z_{1.5}$ of the value function given in equation (C25). After some algebra, I find that investors' objective function at $t=1$ to be:

$$\mathbb{E}_{Z_{1.5}} \left[-exp \left\{ -\alpha \left(A_1 + (Z_{1.5} - \bar{Z}/2) B_1 + (Z_{1.5} - \bar{Z}/2)^2 \frac{C_1}{2} \right) \right\} \right] \quad (\text{C26})$$

With $A_1 = W_1 + \theta_1 \left(\mathbb{E}_{Z_1}(P_2^*) - P_1^* \right) + \frac{\alpha}{2} \sigma^2 (\bar{\theta} + \bar{Z})^2 - \alpha^3 \sigma^4 \kappa_2^2 \bar{Z} (\bar{\theta} - \theta_1 + \bar{Z})$; $B_1 = \alpha \sigma^2 (1 - \alpha^2 \sigma^2 \kappa_2^2) (\bar{\theta} - \theta_1 + \bar{Z})$; $C_1 = \alpha \sigma^2 (1 - \alpha^2 \sigma^2 \kappa_2^2)$

Noting that $(Z_{1.5} - \bar{Z}/2) \sim N(0; \sigma_{Z_{1.5}}^2)$, one can apply lemma 1 in Vayanos and Wang (2012). More precisely, the problem is equivalent to

$$Max_{\theta_1} \mathbb{E}_{Z_{1.5}} \left[-exp \left\{ -\alpha \left(A_1 + B_1 (Z_{1.5} - \bar{Z}/2) \right) \right\} \right] \quad (\text{C27})$$

and replacing $\sigma_{Z_{1.5}}^2$ with $\kappa_{1.5}^2$

We now have a mean-variance problem. One can then obtain the optimal demand function by using the first order conditions, followed by the equilibrium price. I find that

$$P_1^* = \mathbb{E}_{Z_{1.5}}(P_2^*) + \alpha^3 \sigma^4 (\kappa_2^2 + (1 - \alpha^2 \sigma^2 \kappa_2^2) \kappa_{1.5}^2) \bar{Z} \quad (\text{C28})$$

where $\kappa_{1.5}^2 = \frac{\sigma_{Z_{1.5}}^2}{1 + \alpha^2 \sigma^2 (1 - \alpha^2 \sigma^2 \kappa_2^2) \sigma_{Z_{1.5}}^2}$

Note that

$$P_1^* > \mathbb{E}_Z(P_{1.5}^*) > \mathbb{E}_Z(P_2^*) \quad (\text{C29})$$

As a consequence, the implications go through. Indeed, take the proofs of the implications and set $\Delta = \Delta_1 = -\sigma_{Z_{1.5}}^2$. One can also set $\Delta_2 = -\sigma_Z^2$

■